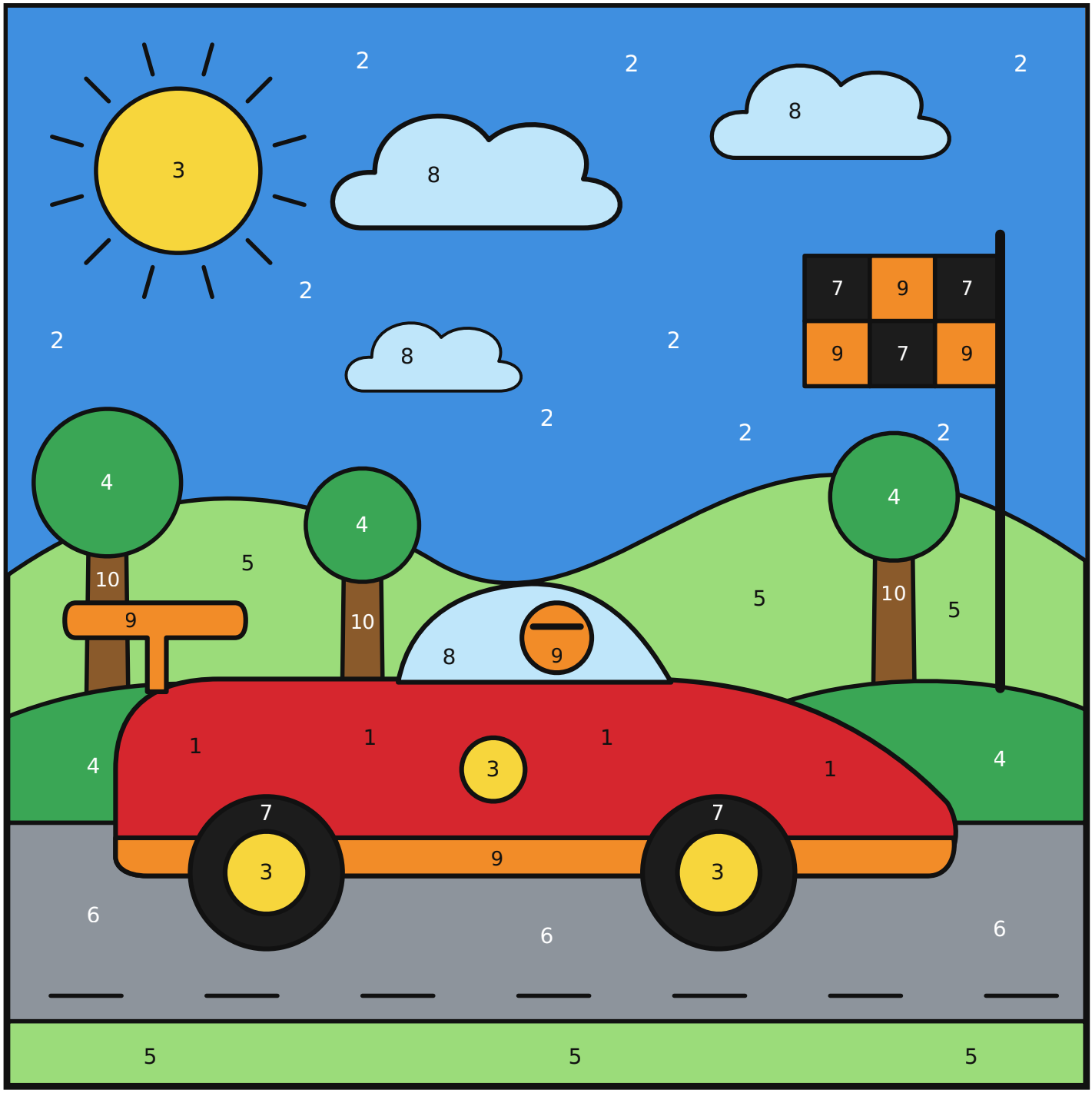
Name: **ANSWER KEY** _____ Date: _____

Quadratics: Color By Number



Color Code:

7 Black $x = 3 \pm \sqrt{5}$	**10** Brown $x = -3$	**3** Yellow $x = 2, -3$	**4** Green $x = -2, 3$	**9** Orange $x = 4, -2$
5 Light Green $x = \pm 4$	**8** Light Blue $(2, -3)$	**2** Blue $x = -2, -3$	**1** Red $x = 2, 3$	**6** Gray $x = \pm 5$

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Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Solve by factoring: $x^2 - 5x + 6 = 0$

$(x - 2)(x - 3) = 0$
$x = 2, 3$

1. Color: **Red**

2.
Solve by factoring: $x^2 + 5x + 6 = 0$

$(x + 2)(x + 3) = 0$
$x = -2, -3$

2. Color: **Blue**

3.
Solve by factoring: $x^2 + x - 6 = 0$

$(x - 2)(x + 3) = 0$
$x = 2, -3$

3. Color: **Yellow**

4.
Solve by factoring: $x^2 - x - 6 = 0$

$(x + 2)(x - 3) = 0$
$x = -2, 3$

4. Color: **Green**

5.
Solve: $x^2 = 16$

take the square root of both sides
$x = \pm 4$

5. Color: **Light Green**

6.
Solve: $2x^2 = 50$

$x^2 = 25$
$x = \pm 5$

6. Color: **Gray**

7.
Use the quadratic formula to solve: $x^2 - 6x + 4 = 0$

$a = 1, b = -6, c = 4$
$x = \frac{6 \pm \sqrt{(36 - 16)}}{2} = \frac{6 \pm 2\sqrt{5}}{2}$
$x = 3 \pm \sqrt{5}$

7. Color: **Black**

8.
Find the vertex of $y = x^2 - 4x + 1$

$x = \frac{-b}{2a} = \frac{4}{2} = 2$
$y = 2^2 - 4(2) + 1 = -3$
(2, −3)

8. Color: **Light Blue**

9.
Solve by factoring: $x^2 - 2x - 8 = 0$

$(x - 4)(x + 2) = 0$
$x = 4, -2$

9. Color: **Orange**

10.
Find the equation of the axis of symmetry of $y = x^2 + 6x + 5$

$x = \frac{-b}{2a} = \frac{-6}{2}$
$x = -3$

10. Color: **Brown**